

1. If a display has a resolution of **1920 × 1080**, calculate the total number of pixels on the screen.
2. An image has resolution 1024 × 768. 24-bits are used for a pixel. Now find out the image size in bytes.
3. A display resolution is 1280 × 720. What will be the aspect ratio of the screen?
4. Two points are given: P1(2,3) and P2(8,9). Find the slope of the line. Write the equation of this line.
5. Using DDA algorithm, Find out the intermediate pixel positions of a line from (1,1) to (5,9).
6. A line is passing from (1,2) to (7,5). Now calculate the intermediate pixel positions using Bresenham's algorithm. Show the calculation for decision parameter at each step.
7. **A circle has a radius 7. Now find out the pixel positions of first octant only using Mid-point circle drawing algorithm.**
8. Or, find out the pixel positions of the circle using Mid-point circle drawing algorithm.
9. Or, find out the pixel positions of first four octant using Mid-point circle drawing algorithm.
10. An ellipse has radius on x-axis is 5 and radius on y-axis is 3.
 - a) Write the equation of the ellipse.
 - b) Find out the length of major axis and minor axis.
 - c) What is the break condition of region Q1?
 - d) Find out the intermediate pixels of Q1 and Q2 of the given ellipse.
11. The equation of a circle is $(x^2 + y^2 = 49)$.
 - a) Find out the radius of the circle.
 - b) Find out the position of (2,-3) pixel about the circle. (inside/outside).